Notice, question, predict

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Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use observations to make a prediction that can be tested.

SUCCESS CRITERIA - I can...

1. Separate supplied facts from statements that have no support.
2. Extend a position pattern and state what your prediction treats as true.
3. Name an observation that could test the prediction.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “evidence” and “prediction” to explain your answer. Sentence starters are optional.

evidence: observations or measurements that support an idea

prediction: what a model says may happen next

model: a drawing, explanation or equation for a physical situation

01 Read a scale

A ruler shows a toy car at **2 cm**, then at **7 cm**.
1. **Draw both positions** and the change.
2. **How far** apart are the marks?

Sentence starter: In my work, I used … to show … . I labeled it … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\Delta x = x_f - x_i$; $x_{next} = x_{now} + \Delta x$.
For a repeating position pattern only. i = initial; f = final. **Check the pattern against actual** measurements.

Worked reasoning: A cart is at **2**, **5** and **8 m** at **0**, **1** and **2 s**. Equal jumps of **3 m** suggest **11 m** at **3 s**, if the pattern continues.

Your turn: A motion sensor recorded the first three positions in the table.
1. **Add the predicted position** at **3 s**.
2. **Draw the four positions** on a line and distinguish the prediction from the measurements.

Sentence starter: I predict … . I used … to make this prediction because … .

Time (s)	Position (m)
0	2
1	5
2	8
3	Predict: ___

Marzano check - circle: ① ② ③

03 Find the useful details

A fictional space report says: "Repeated images show an object approaching Earth. Its path and size remain uncertain." Name two statements the report supports and one conclusion it does not support. **Explain the difference** between an observation and a forecast.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

04 Fit a relationship

A motion sensor tracks a cart at one-second intervals.
1. **Use the table** to **write a position rule**, x(t), and **predict x** at **5 s**.
2. **Explain what you** are treating as true that lets you extend the pattern.

Sentence starter: I predict … . I used … to make this prediction because … .

t (s)	x (m)
0	1
1	4
2	7
3	10

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

Notice, question, predict

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Challenge the prediction

At **5 s**, the sensor measures **14.0 ± 0.2 m**. **Mark this interval** and your prediction from question **4** on the number line. Do they agree? **Give two possible reasons** for any disagreement.

Sentence starter: I predict … . I used … to make this prediction because … .



Position (m); right is positive

Marzano check - circle: ① ② ③

06 Design the observation

A telescope team can make two more observations of the fictional asteroid.
1. **Describe what** to measure and when.
2. **Explain how the pair of** measurements would improve a prediction of its future position.

Sentence starter: I predict … . I used … to make this prediction because … .

Marzano check - circle: ① ② ③

07 Compare two answers

Statement A: “A straight-line calculation predicts where the asteroid may be.”
Statement B: “The asteroid will hit Earth.”
What extra measurements and physics would you need to move from statement A to statement B? Explain.

Sentence starter: I predict … . I used … to make this prediction because … .

Marzano check - circle: ① ② ③

08 Choose useful measurements

Two groups time the same toy car over the same marked track. One records three times; the other writes "pretty fast." Which record would let someone check the statement? Explain.

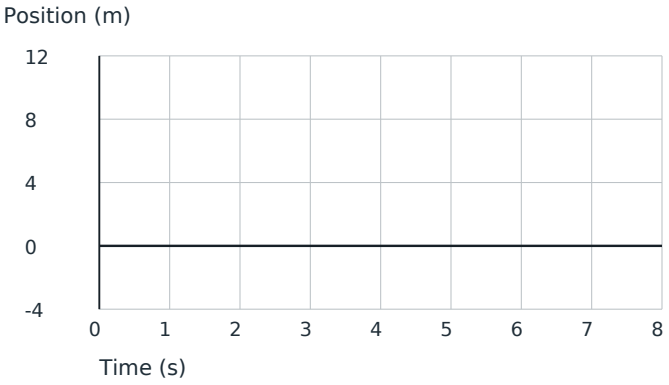
Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: ① ② ③

09 Exit check

A robot's positions are **12**, **8** and **4 m** at **0**, **2** and **4 s**. Plot them, find the velocity, and **predict its position** at **7 s**. **State something you** are treating as true and a later measurement that could test it.

Sentence starter: I predict … . I used … to make this prediction because … .

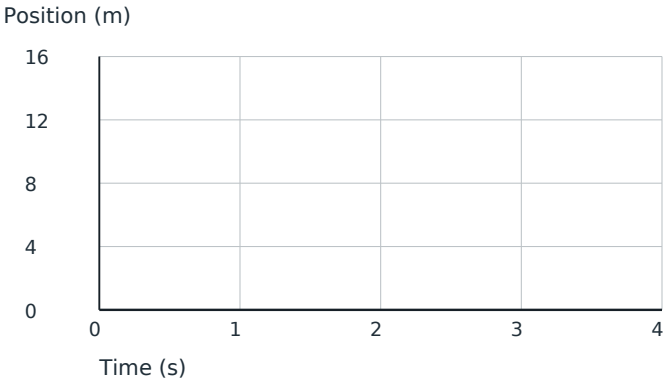


Marzano check - circle: ① ② ③

10 STRETCH: test the prediction

Return to the worked example in problem **2**. Construct two position rules that match all three observations in the worked case but disagree at **4 s**. **Explain why the data** do not guarantee either extrapolation.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



Marzano check - circle: ① ② ③